Department of Computer and Information Systems Engineering

CS-406 Computer Engineering Project

Proposal for the Final Year Design Project

Title	In-Line Tablet Quality Control using Computer Vision

Domain	Domain 1	Domain 2	Domain 3	Domain 4	Domain 5	Domain 6
	Computer Vision & Image Processing	Deep Learning	Data Engineering & MLOps for Model Lifecycle	Human–Machine Interface (HMI) & Dashboarding		

1. Nature of Project [Tick all that applicable]

☒ New Project OR ☐ Extension of Existing Project	☒ Industrial Collaboration	☐ Funded
☐ Other Department Collaboration (If yes) Department Name________________	☐ Other Academic Institution Collaboration (If yes) Institution Name_______________	

2. Brief Outline (*Problem Identification and Significance*)

Conventional tablet quality control (QC) relies on off-line, destructive tests applied on small random samples, which are slow, labour intensive, and may miss defects. This project proposes an in-line, non-destructive, 100% inspection system that uses machine vision and deep learning to (1) detect common surface defects on tablet cores from surface texture. The goal is to check tablet quality by identifying defects at an early stage, enabling timely intervention to improve patient safety and reduce wastage.

Defects targeted: pinholes, spots, capping, chipping

3. Objectives

1. **Design & build** an imaging rig (camera + lighting) mounted on a conveyor.
2. **Develop models** for real-time defect detection/classification and texture-based property prediction using a YOLO-family baseline; benchmark modern alternatives.
3. **Achieve target KPIs:**
    - Defect detection accuracy ≥ **95%**, sensitivity ≥ **95%** on held-out in-house data.
    - End-to-end throughput ≥ **6k tablets/hour** with scalability paths to ≥ **18k/hour** (multi-row conveyor).

4. **Deliver production-grade software**: inference pipeline, HMI dashboard, QC reports, audit logs.

4. Scope

- Imaging of tablet cores on a conveyor with controlled illumination.
- Dataset curation, labeling, training, and evaluation of DL models.
- Web dashboard for live stats, batch summaries, and traceability.

5. Proposed Methodology

A. System Architecture & Mechatronics

- **Conveyor module:** adjustable speed; lanes (1–3) with 3–4 mm spacing between tablets.
- **Imaging module:** 12 MP area-scan RGB camera, diffuser light to accentuate micro-textures and edges.

B. Data & Annotation Plan

- Collect multi-lighting, multi-orientation images for each tablet; capture both faces.
- Curate balanced datasets covering 5 defect classes.
- Label defects (bounding boxes/instances) and class bins using lab-measured ground truth.
- Split: train/validate/test with strict batch/lot separation. Maintain a small blind test set.

C. Modeling & Training

- **Detection:** YOLO-family baseline (e.g., YOLOv5s) training; heavy augmentations (mosaic, flips, HSV, blur, CutOut) tuned for texture robustness.
- **Hyperparameters:** epochs; batch sized to VRAM; cosine LR; label smoothing for robustness.
- **Metrics:** accuracy, class-wise precision/recall, confusion matrices; throughput (fps), latency budget (< 25 ms/frame).

D. Real-Time Pipeline & HMI

- **Capture** → **Preprocess** → **Inference** → **Post-process** → **Actuate** (detect) → **Log**.
- HMI: defect heat-maps, per-class counts, batch reports (PDF/CSV), audit trail (time-stamped, operator ID).

Deliverables:

- Literature Review and Requirements
- Data Annotation
- Data Preprocessing and Augmentation
- Model Training (Defect)
- Model Training (Properties)
- Real-time Pipeline & HMI
- Rig Design (Conveyor, Lighting)
- Validation & KPI Hit
- Report, Demo & Handover

6. Resources Involved

Hardware

- Scan camera (≈12 MP), LED light, conveyor belt with variable speed.

Software

- Python, PyTorch, YOLO toolchain,TensorRT, OpenCV; LabelImg/Roboflow; FastAPI; InfluxDB + Grafana/Streamlit dashboard; Git + DVC for data versioning.

7. Description of Industrial Support (If any)

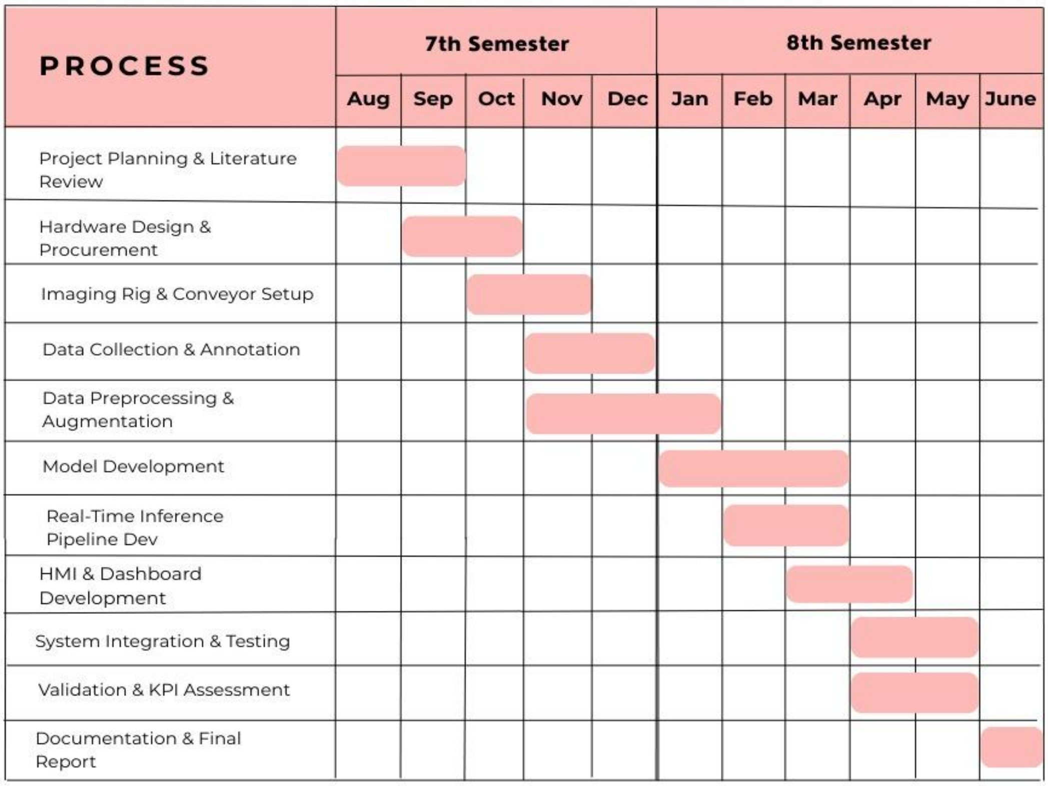
Dataset is provided by a contact person from the pharmaceutical industry. Furthermore,

Nutrasource International extends collaborative support in the form of technical feedback, domain expertise, and representative guidance.

8. SDGs (If Applicable)

☐ No Poverty	☐ Zero Hunger
☒ Good Health and Well-Being	☐Quality Education
☐ Gender Equality	☐ Clean water and Sanitation
☐ Affordable and Clean Energy	☐ Decent Work and Economic growth
☒ Industry, Innovations and Infrastructure	☐ Reduced Inequalities
☐ Sustainable Cities and Communities	☒ Responsible Consumption and Production
☐ Climate action	☐ Life Below Water
☐ Life on Land	☐ Peace, Justice and Strong Institutions
☐ Partnerships	

9. Gantt Chart



10. Details of Project Team

i. Students

No.	Name	Seat No.	Signature (s)
1	Manahil Ejaz	CS-22011	
2	Mahnoor Zia	CS-22101	
3	Anoosha Khalid	CS-22104	
4	Laiba Iqrar	CS-22112	

ii. Supervisors / Advisors

	Name	Designation & Department	Address & Contact	Signature(s)
Supervisor	Syed Zaffar Qasim	Assistant Professor, CIS		
Co-Supervisor (If any)				

Industrial Advisor (If any)	Syed Khawaja Karimullah	Chief Executive Officer		

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Project Serial No.: ___________ Dated: Sept 25, 2025________	Signature Convener Steering Committee	Signature FYP Coordinator

☐ Proposal Approved	☐ Not Approved	☐ Returned for Clarification / Modification
Comments: (if any)		

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(Signature of Chairperson)

Date: _______________________________